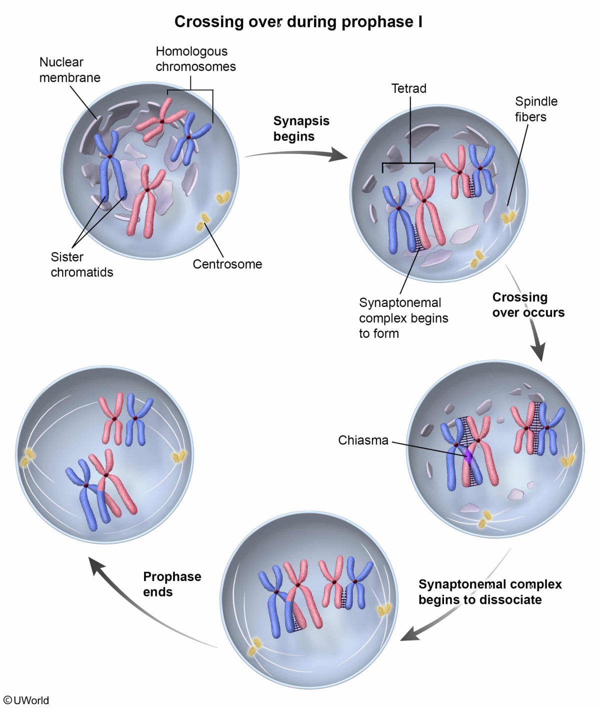
Which of the following cellular processes would most likely lead to increased genetic variation in a population of organisms?

- A. DNA replication during S phase of the cell cycle [2%]
- B. Synapsis involving two homologous chromosomes [42%]
- C. Alternative splicing of gene transcripts [52%]
- D. Cell division via mitosis [2%]

Correct
42% answered correctly

Explanation:



In eukaryotes, reproductive cells (gametes) are produced via meiosis, during which a diploid (2*n*) parent cell divides to produce four genetically distinct haploid (*n*) daughter cells. Meiosis has four phases (like mitosis) but these phases occur in two successive cell division stages (meiosis I and II).

In **prophase I** of meiosis, **homologous chromosomes** line up side-by-side during **synapsis**. Because each homologous chromosome consists of two identical (sister) chromatids, this adjacent chromosomal alignment forms a tetrad (ie, four chromatids total). Tetrads arise when a *synaptonemal complex* (protein structure) forms between homologous chromosomes and holds them together tightly.

The tetrad structure allows physical contact between the paternal and maternal chromosomes at the *chiasma*, a point where a chromosome segment can break off and rejoin with the opposite homologous chromosome. This exchange of DNA is the hallmark feature of the process called *crossing over*.

Crossover events result in **daughter cells** with chromosomes containing **combinations of alleles** that differ from those in the **parent cell**, leading to eukaryotic genetic recombination and **increased genetic variation**. Because crossover events produce genetically unique gametes, the offspring that develops after fertilization is genetically different from either parent, leading to increased genetic variation.

(Choice A) DNA replication *does not* involve the exchange of genetic information as during crossover events in meiosis. DNA replication (synthesis) occurs during S phase of the cell cycle and precedes mitosis. Because of proofreading, DNA replication is generally unlikely to introduce errors (ie, mutations), and is therefore a *less likely* source of genetic variation than crossing over.

(Choice C) In alternative splicing, a single gene transcript is processed to produce various mRNA molecules that encode different proteins, depending on the inclusion or exclusion of particular exons. Alternative splicing increases *protein* diversity but does not increase *genetic* variation because the genome is not altered.

(Choice D) Mitosis is the process by which a cell divides to create two *genetically identical* daughter cells. During mitosis, homologous chromosomes do *not* pair up into tetrads, and crossing over does *not* occur.

Educational objective:
In eukaryotes, genetic recombination occurs via crossover events (ie, exchange of DNA segments between homologous chromosomes). Synapsis, the joining of homologous chromosomes into tetrads, occurs during prophase I of meiosis and is required for crossing over to occur. Crossover events increase genetic variation by mixing maternal and paternal alleles into a single chromosome that is then inherited by the offspring.

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Subject:	**System:**
Biology	Genetics and Evolution